

Opportunities in XR Simulations

Strategic Intelligence Report | 2024–2030

1. Overview & Scope

1.1 What Are XR Simulations?

Extended Reality (XR) is an umbrella term encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR). In the context of simulations, XR refers to the use of these immersive technologies to replicate real-world environments, processes, and scenarios for training, analysis, planning, or operational support.

- VR → Fully immersive, closed digital environments (e.g., surgical rehearsal, combat training)
- AR → Digital overlays on the real world (e.g., industrial maintenance guidance, AR HUDs)
- MR → Dynamic blending of real and virtual — interactive and context-aware (e.g., digital twins, collaborative design)

1.2 Core Categories

Category	Description
Training Simulations	Enterprise workforce upskilling, onboarding, soft & hard skills
Digital Twins	Real-time virtual replicas of physical assets & systems
Medical Simulations	Surgical planning, clinical training, therapy & rehabilitation
Defense / Military	Combat training, mission rehearsal, situational awareness

1.3 Why Now? — Key Drivers & Catalysts

Technology Enablers:

- Convergence of 5G, edge computing, and improving hardware form factors
- MR growing at the highest XR CAGR — context-aware, interactive experiences
- OpenXR standardization supported by 13+ major vendors, lowering cross-device development costs

Market Catalysts:

- Market pivoting from gaming-centric to professional productivity and ROI-driven enterprise applications
- Over the past two to three years, AI integration with XR has become a primary focus for next-gen simulations
- The market is primarily driven by significant advancements in VR & AR technologies, combined with growing demand in defense, healthcare, and industrial sectors

2. Market Size & Growth

2.1 Global XR Simulation Market Size (2024–2030)

Source	2024 Baseline	2030 Projection	CAGR
Grand View Research	\$16.4B(immersive training)	\$69.6B	28.3%
MarketsandMarkets	\$24.4B	\$84.9B (2029)	28.3%
ABI Research	\$44.7B	\$299.3B	~37%
Market.us / Scoop	\$62.9B	\$294.6B	30.8%

- The global immersive training market was estimated at USD \$16.4 billion in 2024 and is projected to reach USD \$69.6 billion by 2030, growing at a CAGR of 28.3%. [1]
- The industrial metaverse market, blending digital twins, collaborative design, and immersive training, is projected to reach \$170 billion by 2030 at 37% CAGR. [2]

Consensus Range: \$250B–\$300B (full XR) by 2030 | CAGR: 28–37%

2.2 Breakdown by Sub-Segment

- ♦ **Training Simulations**
 - Fortune 500 adoption now exceeds 75%, with large enterprises three times more likely than mid-size peers to rely on VR for mission-critical training. [2]
 - Organizations that scale VR training beyond 3,000 learners achieve 52% cost savings versus classroom delivery, with training time reductions of 60–75% and safety incident reductions of 35–50%. [2]
- ♦ **Digital Twins**
 - Investments in industrial AI and physical AI are driving increased demand for digital twins — these physically accurate virtual replicas aren't just helping manufacturers streamline planning; they serve as training grounds for AI agents, autonomous vehicles, and robot fleets. [3]
- ♦ **Medical / Healthcare XR**
 - Healthcare is the fastest-growing XR sub-segment, driven by FDA adoption and clinical validation
 - XR technologies are improving patient recovery through immersive care; VR-based rehabilitation tools from companies like MindMaze and XRHealth assist stroke patients, while AppliedVR helps reduce chronic pain and anxiety. [4]
- ♦ **Defense / Military**
 - The Defense AR/VR & Simulation Market will surpass \$10.50 billion in 2025, driven by military demand for AR systems enhancing soldier situational awareness and mission command. [5]

2.3 CAGR & Growth Summary

Sub-Segment	Est. CAGR	Growth Driver
Medical / Healthcare XR	~43%	FDA approvals, surgical training
AR Technology (overall)	~43%	Industrial, defense, enterprise
MR Technology	Highest	Context-aware enterprise workflows
Defense Simulations	High / Steady	IVAS program, procurement mandates
Digital Twins / Industrial AR	~37%	IoT integration, physical AI
VR Training (Enterprise)	~28–31%	Workforce upskilling at scale

3. Training Simulations

3.1 Key Players

- Strivr — With over one million trained in VR, Strivr transforms the employee journey through immersive learning; incubated at Stanford, the platform powers the creation, delivery, management, and measurement of VR-based learning. [6]
- Talespin — Cornerstone's 2024 acquisition of Talespin was a major move, significantly boosting capabilities in soft skills training. [7]
- Mursion — Mursion excels in "human-in-the-loop" simulations providing real-time avatar responses for nuanced soft skills like leadership coaching, offering unparalleled authenticity in conversational practice. [7]
- Moth+Flame — Specialized in military and first-responder immersive training; known for rapid VR deployment at scale

3.2 Enterprise Use Cases

Company	Application
Walmart	Partnered with Strivr to enhance customer service training, transporting employees into virtual store environments. [8]
Coca-Cola	Partnered with Mursion to revolutionize leadership development, giving leaders opportunities to practice decision-making and interpersonal skills in simulated scenarios. [8]
Boeing	Uses VR to train employees on assembly, leading to reduction in training time and costs. [9]
PwC / Accenture / Deloitte	Talespin's CoPilot platform uses AI-powered virtual humans for leadership, communication, and empathy training. [2]

3.3 Effectiveness Data

- Research by the University of Maryland found that VR-based training improves skill acquisition by 28% and accelerates learning up to four times faster than legacy training methods. [9]
- PwC reported that VR-trained employees are 3.75× more emotionally connected to content and 275% more confident applying skills. [2]
- A Deloitte case study showed VR training led to a significant reduction in training costs by approximately 30%. [9]
- According to Strivr, 70% of employees who took part in the VR training programme performed better on their exams than those in standard training, while satisfaction levels were 30% higher. [10]
- One client condensed an entire day of safety training into just 15 minutes in VR, and 97% of trainees felt prepared for real danger. [11]

3.4 Soft Skills vs. Hard Skills Training

Dimension	Soft Skills	Hard Skills
Best Platform	Mursion, Talespin CoPilot	Strivr, Interplay Learning
Use Cases	Leadership, empathy, conflict resolution	Equipment operation, safety, assembly
Approach	Human-in-the-loop avatars, AI NPCs	Procedural simulation, step-by-step VR
Key Metric	Behavioral change, confidence scores	Time-to-proficiency, error rates

- VR provides the opportunity to cultivate soft skills — like conflict resolution and team management — that traditional methods often overlook; behavioral responses induced by VR can mirror those in actual work situations, making skill transfer more seamless. [\[9\]](#)

4. Digital Twins & Industrial AR

4.1 What Are Digital Twins in XR?

Digital twins are real-time, physics-accurate virtual replicas of physical assets, systems, or facilities, rendered and interacted with through AR/MR interfaces. They enable remote monitoring, scenario planning, virtual commissioning, and AI training — all before physical deployment.

- Inputs: IoT sensor feeds, CAD data, operational telemetry
- Outputs: Maintenance alerts, simulation results, optimization recommendations, training environments

4.2 Key Players

- NVIDIA Omniverse — A collection of libraries and microservices for developing physical AI applications such as industrial digital twins and robotics simulation, leveraging NVIDIA's expertise in accelerated computing and AI. [12]
- Siemens + NVIDIA — Utilizing the integration between Siemens Xcelerator and NVIDIA Omniverse, the technology supports a new highly realistic digital twin offering that brings together 3D visualization, simulation, and factory data into one unified, immersive environment.[13]
- PTC / Vuforia — Enterprise AR leader for industrial work instructions and remote assistance
- Unity Industrial — Real-time 3D engine powering interactive digital twin visualization

4.3 Use Cases by Industry

Industry	Application	Example
Manufacturing	Factory layout planning, virtual commissioning	BMW uses NVIDIA Omniverse to create digital twins of its factories, enabling real-time monitoring and optimization of production lines to identify potential bottlenecks before they impact production. [14]
Energy	Plant operations, predictive maintenance	Siemens Energy creates digital twins of power plants using NVIDIA Omniverse, providing a comprehensive view of plant operations for predictive maintenance and optimization of energy output. [14]
Automotive	Vehicle design & testing	Mercedes-Benz uses NVIDIA Omniverse to create digital twins of their vehicles, allowing virtual testing of new designs and reducing the need for physical prototypes. [14]
Robotics Logistics	/ Robotic training without physical hardware	Amazon uses digital twins to train robotic arms to recognize, inspect, and handle new devices — robotic actions configured for assembly, testing, packaging, and auditing entirely in simulation. [3]

4.4 Integration with IoT & Real-Time Data

- Once the real facility is built, virtual assets can connect to real-world sensors, turning the digital twin into a live operational dashboard for day-to-day factory management. [\[15\]](#)
- NVIDIA's Omniverse Blueprint includes physics-AI frameworks and interactive rendering to achieve 1,200× faster simulations and real-time visualization. [\[16\]](#)
- NVIDIA Omniverse has over 300,000 downloads and 252+ enterprise deployments across manufacturing, automotive, robotics, and media industries as of August 2025. [\[17\]](#)

5. Medical Simulations

5.1 Surgical Planning & Rehearsal

- Medivis / Proximie / Surgical Theater — AR platforms for pre-operative visualization and remote surgical guidance
- Proximie enables remote surgical assistance, allowing experts worldwide to virtually "scrub in" and support live operations through AR overlays and shared 3D perspectives. [4]
- Sira Medical gained FDA clearance in February 2024 for its AR preoperative surgical planning application that creates patient-specific, high-fidelity 3D holograms, helping patients understand anatomy and procedure better. [18]

5.2 Medical Training

- Osso VR — A surgical training and assessment platform offering medical companies and healthcare professionals a new way to share knowledge, equipped with one of the world's largest VR training libraries in healthcare. [19]
 - Clinical studies show Osso VR users demonstrate a 230% improvement in overall performance versus traditional training. [20]
- FundamentalVR — At the heart of the system is the Surgical Haptic Intelligence solution, calibrated to mimic the real-life sensations of tissue variants and medical tools. [19]
 - FundamentalVR tracks performance metrics such as force applied, procedural accuracy, and efficiency, with students receiving detailed feedback after every session. [21]
- Oxford Medical Simulation — AI-driven clinical decision simulation for medical students and residents

5.3 Therapy & Rehabilitation

- VR-based rehabilitation tools such as those from MindMaze and XRHealth assist stroke and trauma patients in regaining motor skills through gamified exercises. [4]
- Platforms like AppliedVR have demonstrated up to 69% reductions in pain intensity during VR sessions and improved emotional well-being post-treatment. [4]
- Mental health programs are also leveraging XR to treat PTSD, phobias, and depression through controlled exposure and guided therapy environments. [4]

5.4 FDA Approvals & Clinical Validation

- The FDA released a list of 69 clinical VR and AR products granted clearance since 2015, with a third of those approvals granted in just the last 18 months. [18]
- AppliedVR's EaseVRx received FDA Breakthrough Device designation for treating treatment-resistant fibromyalgia and chronic intractable lower back pain — one of the first VR digital therapeutics to achieve this. [22]
- Over the past five years, interest in VR at the intersection of orthopedic surgery education has grown significantly; its popularity is attributed to its convenience, cost, and depth of educational experience given real-time performance metrics. [23]
- VR training programs achieve positive ROI within 18–24 months in healthcare settings. [20]

6. Defense & Military XR

6.1 Key Programs

Microsoft IVAS / HoloLens (U.S. Army)

- Based on Microsoft's commercial HoloLens technology, IVAS is meant to replace the Army's current night vision and Nett Warrior platforms; the Army awarded Microsoft a 10-year contract worth up to \$21.9 billion for IVAS in March 2021. [\[24\]](#)
- Version 1.2 introduced an embedded "Squad Immersive Virtual Trainer" that projects holographic and mixed reality imagery to train soldiers in combat, with navigation applications displaying maps and positions of friendly and enemy forces. [\[25\]](#)
- From February 2025, Anduril Industries partnered with Microsoft to assume oversight of production, future hardware/software development, and delivery timelines. [\[25\]](#)

6.2 Key Contractors & Players

Player	Role
Microsoft / Anduril	IVAS program — AR HUD for Army close combat forces
Varjo	Military-grade VR/XR headsets and solutions for mission-critical training and simulation across air, land, and sea. [26]
Red 6	AR air combat training system (ATARS) for live airborne dogfight simulation
CESI	Defense simulation systems integrator; live-virtual-constructive environments

- The U.S. Army adopted Varjo's XR-3 headsets for its Reconfigurable Virtual Collective Trainer program, allowing helicopter crews to train together using simulated controls and immersive visuals and rehearse missions in realistic conditions without full-scale aircraft simulators. [\[27\]](#)

6.3 Use Cases

- Combat Training — IVAS Squad Immersive Virtual Trainer projects holographic imagery via headset to train soldiers in combat scenarios. [\[25\]](#)
- Mission Rehearsal — Navigation applications allow users to lay out 3D maps of terrain, follow compass points in the HUD, and identify friendly/enemy locations; the device can load a "sand table" or internal map of a shoot house for tactical scenarios. [\[28\]](#)
- Flight / Pilot Training — Embry-Riddle University found that adding VR cut time to first solo flight by 30%; the U.S. Air Force reports that after VR adoption, students reached solo ~10 flights earlier than the traditional approach. [\[29\]](#)
- Maritime — Teams practice navigation, watchkeeping, and emergency response in high-fidelity XR maritime environments

6.4 Budget & Procurement Data

- For fiscal year 2025, the Army requested \$255 million procurement for 3,162 IVAS 1.2 HUDs, \$58.6 million for HUD R&D, and \$18.9 million for Squad Immersive Virtual Trainer (SIVT), test and evaluation funding for IVAS. [\[24\]](#)
- The IVAS program could be worth as much as \$21.9 billion if it comes to full fruition. [\[30\]](#)

9D Technologies | AI Research



- The Defense AR/VR & Simulation Market surpassed \$10.50 billion in 2025, with strong revenue growth through to 2035. [\[5\]](#)
- Armed forces now view AR headsets not as point gadgets but as persistent mission platforms that must interoperate with drones, sensors, and C2 networks — procurement and R&D budgets are following that system's view. [\[5\]](#)

7. Competitive Landscape Matrix

Player	Sub-Segment	Core Technology	Traction	Funding / Scale
Cornerstone	Enterprise Learning	LMS + Immersive VR	Global enterprise clients	Acquired Talespin (2024)
Mursion	Soft Skills Training	Human-in-loop AI Avatars	35+ industries; Coca-Cola, J&J	Acquired Valor Performance (2024)
NVIDIA Omniverse	Digital Twins	OpenUSD, Physics AI, RTX	300K+ downloads; 252+ enterprises	NVIDIA (\$3T+ market cap platform)
Siemens Xcelerator	Industrial Digital Twins	CAD + Omniverse integration	BMW, energy sector, auto OEMs	Enterprise SaaS / Global
Osso VR	Medical Training	Surgical VR + Haptics	230% performance improvement data	Meta preferred partner
FundamentalVR	Surgical Simulation	HapticVR + Procedural Rehearsal	Amazon, Bayer, UCL Hospital	Growth-stage; MedTech focus
AppliedVR	Therapy / Rehab	Digital Therapeutics VR	60K+ patients, 200 hospitals	FDA Breakthrough Device designation
Microsoft / Anduril (IVAS)	Defense / Military	AR HUD + Edge AI	US Army; \$21.9B contract ceiling	DoD funded
Varjo	Defense Simulation	Military-grade MR Headsets	U.S. Army RVCT program	Hardware/SaaS; defense contracts
Red 6	Airborne Combat Training	AR Dogfight Simulation	USAF programs	Defense venture-backed

8. Gaps & Whitespace

8.1 What's Missing in Current Solutions?

- Fragmented ecosystems — Most platforms are siloed by vertical; no unified cross-domain XR simulation OS
- Hardware dependency — Many enterprise solutions tied to specific headset vendors, limiting scalability
- Content creation bottleneck — Building high-fidelity simulation content is time-intensive and expensive; no-code authoring remains nascent
- Assessment gaps — Emotional and cognitive state measurement in training is still immature; current analytics focus on task completion over true behavioral change
- Interoperability — Limited ability to port simulations across VR, AR, and MR devices without redevelopment
- Accessibility — High upfront costs (\$50K-\$200K setup in medical) remain a barrier for mid-market adoption

8.2 Role of AI in Next-Gen Simulations

- Generative AI → Auto-generates simulation environments, scenarios, and NPC behaviors at scale; eliminates manual content creation bottleneck
- AI NPCs / Virtual Humans → Cornerstone On Demand unveiled Immerse Companion in October 2024, a generative AI solution providing real-time interactions with virtual human companions for role-play, coaching, and skill-building sessions on demand. [1]
- Adaptive Learning → AI adjusts difficulty, pacing, and scenario complexity in real-time based on learner biometrics and performance signals
- Physics AI → NVIDIA's Omniverse Blueprint uses physics-AI frameworks achieving 1,200× faster simulations [16], enabling real-time responsive environments
- Predictive Analytics → With Strivr, enterprises gain unique learning and assessment data to measure training effectiveness, evaluate role potential, and predict learning outcomes at scale. [6]

8.3 Where Could a Spatial AI Assistant Add Value?

Gap	Spatial AI Opportunity
On-demand scenario generation	AI that creates dynamic, context-aware training environments from a simple prompt
Real-time coaching	In-simulation AI coach that observes, corrects, and guides learners without breaking immersion
Cross-platform delivery	AI layer that adapts content intelligently across VR headset, AR glasses, or 2D screen
Knowledge retrieval at point of work	AR overlay that surfaces SOPs, schematics, or expert guidance during physical tasks
Simulation debrief intelligence	Post-session AI that synthesizes behavioral data into actionable development recommendations

9. Key Takeaways

1. The market is large and accelerating — The full XR simulation market is on track for \$250B–\$300B by 2030 at a 28–37% CAGR, with healthcare and defense as the fastest-growing verticals.
2. Training simulations have proven ROI — Organizations scaling VR training beyond 3,000 learners achieve 52% cost savings, 60–75% training time reductions, and 35–50% fewer safety incidents. [\[2\]](#)
3. Digital twins are becoming operational infrastructure — "Digital twins have become essential in the age of industrial AI, enabling the simulation and optimization of entire production lines and training robotics virtually before a single piece of hardware is installed." [\[13\]](#)
4. Medical XR is gaining regulatory legitimacy — The FDA has released a list of 69 clinical VR and AR products granted clearance since 2015 [\[18\]](#), and the pace of approvals is accelerating.
5. Defense is the largest single-contract opportunity — The IVAS program alone could be worth as much as \$21.9 billion [\[30\]](#) and is shifting toward AI-integrated, multi-sensor platforms.
6. AI is the next unlock — Generative AI for content creation, adaptive simulation, and intelligent coaching represents the biggest whitespace opportunity across all four sub-segments.
7. Spatial AI Assistants are the missing layer — No current platform meaningfully integrates a persistent, intelligent, cross-modal AI agent that can guide, adapt, assess, and coach inside live simulation environments in real time.

Sources & Data References

Report compiled using data from Grand View Research, MarketsandMarkets, ABI Research, NVIDIA, DefenseScoop, FDA, PubMed, and industry primary sources. Data current as of Q1 2026.